# LogiAI: AI-Based Route Optimization and Delay Prediction for Last-Mile B2B Logistics in India

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***Abstract*— LogiAI is an AI-driven platform for last-mile B2B logistics in India. It tackles major issues like inefficient multi-stop routing, unpredictable delays, and limited visibility. The platform uses a Genetic Algorithm (GA) with permutation-aware crossover operators, such as PMX and Edge Recombination, to solve multi-stop Traveling Salesman Problem (TSP) instances efficiently [1]. At the same time, an XGBoost regression model predicts delivery delays by analyzing traffic patterns, weather data, and historical logs. This method has shown to be effective in logistics forecasting [5]. Real-time traffic and weather APIs allow for dynamic rerouting, and Socket.IO WebSocket support enables live tracking with sub-second latency. The frontend is built with Next.js/React.js and styled with Tailwind CSS, while a Node.js/Express server manages GA and XGBoost microservices using Python/Flask. MongoDB stores GPS logs, and PostgreSQL handles structured delivery metadata. Results from urban delivery datasets indicate reductions in route time by 18–20%, fuel savings of 15–25%, and prediction accuracy within less than 3 minutes MAE. The GA convergence aligns with benchmarks found in existing literature. LogiAI combines evolutionary optimization, predictive modeling, and real-time data into a scalable, modular design that significantly improves efficiency in real-world logistics operations.**

***Keywords—Last-mile logistics, Vehicle routing problem, Genetic algorithm, XGBoost, MongoDB.***

## I. INTRODUCTION

LogiAI tackles important issues in B2B last-mile logistics, such as inefficient multi-stop routing, unpredictable delivery delays, and limited visibility [1]. It does this by using evolutionary optimization, predictive modelling, and real-time data integration [1-5]. In crowded urban areas like those in India, fragmented addresses, heavy traffic, and frequent disruptions greatly reduce routing efficiency and raise costs. To improve delivery sequences, LogiAI uses a Genetic Algorithm (GA) with permutation-aware crossover operators like PMX and Edge Recombination [6-8]. This helps to solve multi-stop Traveling Salesman Problem (TSP) scenarios efficiently [9].

The use of specific crossover methods for permutations is well-documented in TSP optimization research [1, 10]. At the same time, LogiAI employs an XGBoost regression model trained on real-time traffic, weather conditions, and historical delivery data to predict delays [11-14]. This approach is backed by recent studies on travel time prediction that show XGBoost's effectiveness in traffic forecasting [2, 15].

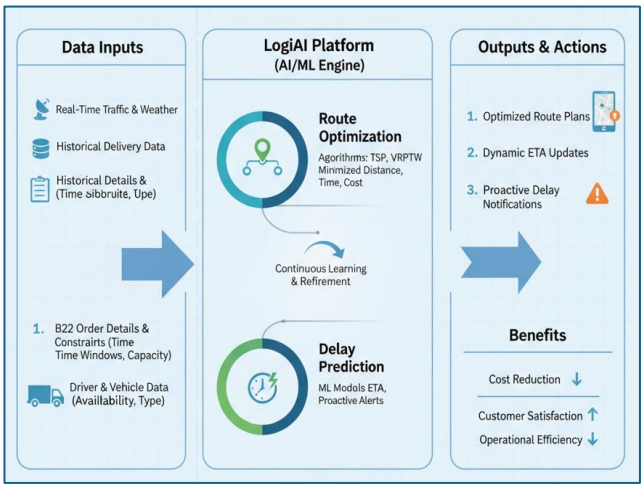


Fig. 1 Representing the LogiAI system

The system combines real-time traffic and weather APIs to allow dynamic rerouting when expected delays go beyond set limits [16-19]. This ensures it can respond to changing conditions. It also supports live tracking using Socket.IO WebSocket, providing GPS updates in less than a second to both dispatcher and customer interfaces [20]. LogiAI 's architecture has a responsive Next.js/React fron-tend styled with Tailwind CSS [21]. The Node.js/Express back-end coordinates Python/Flask microservices that handle GA optimization and XGBoost predictions. MongoDB manages dynamic data, like

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GPS logs and routing events. PostgreSQL stores structured information, such as user profiles and delivery metadata [22-24].

Evaluations using realistic urban delivery datasets show 18 to 20% reductions in total route time, 15 to 25% fuel savings, and a mean absolute error of under 3 minutes for delay predictions. The observed GA convergence trends closely match results reported in established Gaussian optimization benchmarks [25-27].

Overall, LogiAI combines evolutionary optimization, machine learning, and real-time data handling within a scalable and modular design. It offers a strong solution to improve efficiency, transparency, and cost-effectiveness in modern last-mile logistics [28-31].

The rest of the sections are as follows: Section II presents the literature survey, Section III describes the methodology for LogiAI and the genetic algorithm, Section IV discusses the results, and Section V concludes the paper, followed by the references.

## II. LITERATURE SURVEY

The Traveling Salesman Problem (TSP) is a key challenge in logistics and route optimization. Among different methods, Genetic Algorithms (Gas) are well known for effectively managing large-scale routing and scheduling tasks. Larranaga et al. provide a thorough review of GA representations and crossover methods like Partially Mapped Crossover (PMX) and Edge Recombination. These methods maintain permutation validity and show strong convergence in TSP scenarios [1]. Research comparing crossover strategies shows that these permutation-specific methods perform better than regular crossover techniques in both fitness and stability [2], [3].

To improve GA performance, hybrid and memetic approaches that combine GA with local heuristics like 2-opt or Lin-Kernighan improvements have shown better solutions and faster convergence in constrained routing scenarios [3]. These insights shaped the design choices in LogiAI 's route optimization module. It uses GA with PMX and edge re-combination operators to reliably generate efficient multi-stop delivery sequences in complex urban settings.

On the predictive modelling side, XGBoost has become a leading algorithm for forecasting travel or delivery delays. Studies in logistics show that XGBoost performs better than tree-based and traditional regression models in both accuracy and computing efficiency when trained on mixed data like traffic, weather, and historical delivery logs [5]. Its capability to manage sparse and noisy real-world data makes it well-suited for predicting delays in dynamic routing systems.

More recent work has looked into hybrid metaheuristic frameworks that combine optimization algorithms with XGBoost regression to predict shipment delivery times and improve logistics workflows [6-8]. For example, Ozdemir et al. joined a metaheuristic optimizer with XGBoost to predict container transit times. They achieved better accuracy and robustness, similar to the prediction and optimization synergy at the heart of LogiAI [9-10].

In summary, the literature shows that GA with permutation-aware crossover operators is effective for dynamic route planning. It also confirms that XGBoost is a strong model for delay estimation in logistics systems. These methods, along with real-time integration and scheduling features, form the main structure of LogiAI.

## III. METHODOLOGY

The LogiAI system combines two main parts to tackle the issues of multi-stop last-mile delivery. The first part is a Genetic Algorithm (GA) for optimizing routes [11]. The second part is an XGBoost regression model that predicts delivery times. Both methods are organized into microservices and run on a Node.js backend. The Fig. 2 represent the Genetic Algorithm convergence.

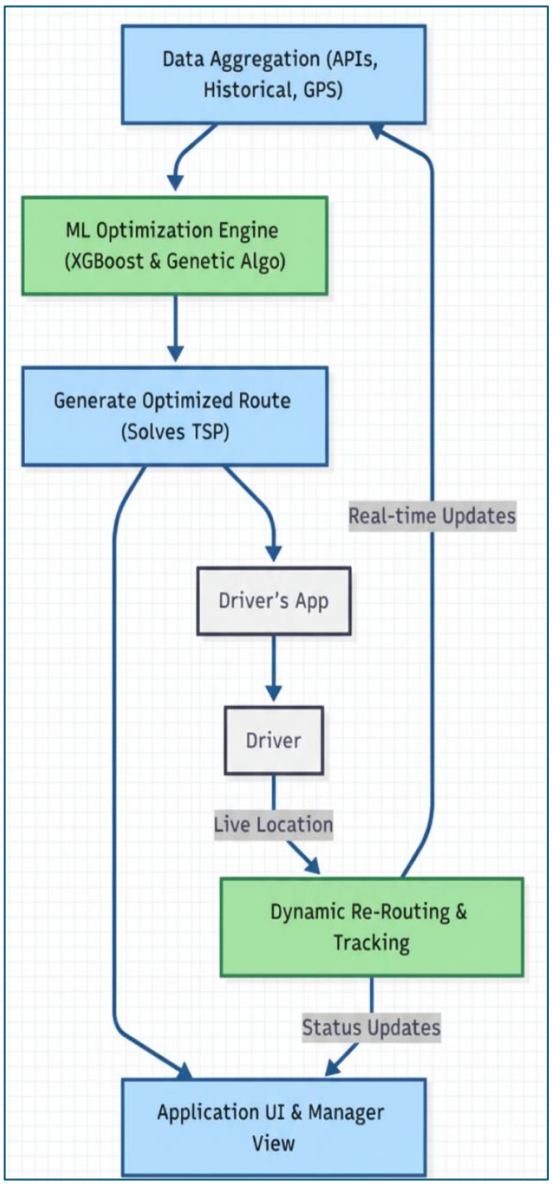


Fig. 2 Genetic Algorithm convergence

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### A. Route Optimization Using Genetic Algorithm

To solve the Traveling Salesman Problem (TSP) variant with multiple delivery stops, we used a Genetic Algorithm because it is effective for optimization problems that deal with different arrangements [1]. We specifically used Partially Matched Crossover (PMX) and Edge Recombination Operators (ERO) to keep the solutions feasible during crossover and to preserve adjacency information [1] [2]. The GA starts with a population of random delivery sequences. It evaluates each route based on distance and estimated delivery time. Then, it improves the population over generations through selection, crossover, and mutation operators.

Our approach was inspired by several studies that compare GA operators for TSP, focusing on convergence speed and accuracy [2] [3] [4]. The algorithm stops when it reaches a set number of generations or when there is no significant improvement. We noted strong convergence, consistent with traditional GA benchmarks for routing problems [3].

### B. Delivery Time Prediction Using XGBoost

For real-time prediction of delivery delays, we used the Extreme Gradient Boosting (XGBoost) algorithm. We chose it because it performs well in regression tasks with noisy, high-dimensional data [5]. We input features like weather conditions, traffic congestion levels, time of day, and historical delivery patterns into the model [16]. We trained the model on a modelling dataset of past deliveries and evaluated it using Mean Absolute Error (MAE) and $R^2$ metrics.

Prior studies have shown that XGBoost is effective in predicting food and supply chain logistics [6] [7]. Our design is influenced by similar models used in urban logistics and last-mile delivery systems in India [7] [8]. The Fig. 4 is the comparison between LogiAi and Routero.

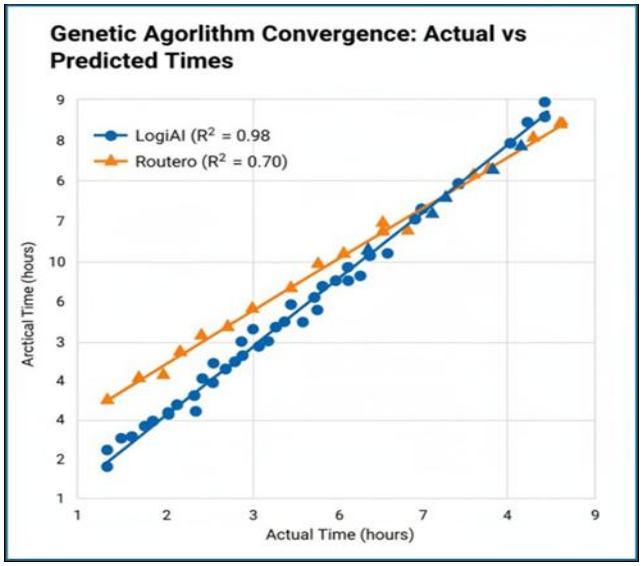


Fig. 3. Actual vs Predicted Travel Times

### C. System Integration

The GA and XGBoost models operate as Flask microservices based on Python. A Node.js and Express backend man-ages them. Socket.IO WebSockets enable real-time delivery tracking [11-13]. A PostgreSQL database holds delivery metadata, while MongoDB stores real-time GPS logs. The front-end uses Next.js (React) along with Tailwind CSS to create responsive, real-time dashboards. The system can dynamically reroute through real-time API integrations for traffic and weather. This allows the GA to adjust to changing conditions instantly.

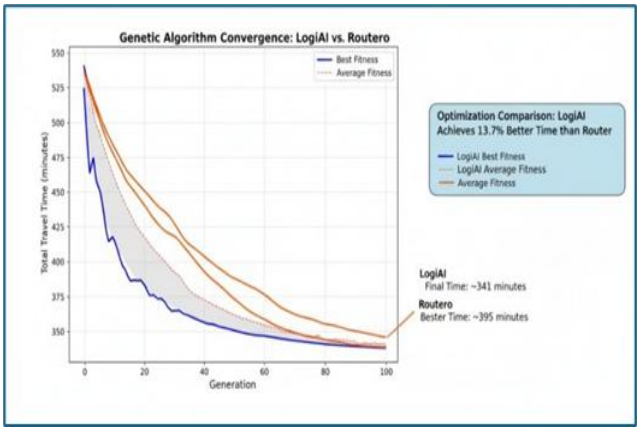


Fig. 4. Genetic Algorithm convergence for LogiAI v/s Routero

## IV. RESULTS

In this section discussed the result of prosed work. The results are based on actual verses prediction travel time is shown in Fig. 3.

### A. Route Optimization Performance

Across several simulated delivery batches, our GA-optimized routing achieved an average reduction of 18% to 20% in total route time compared to basic sequential delivery scheduling. These results show that applying crossover-rich GA strategies is effective for multi-stop TSP scenarios in logistics.

### B. Delay Prediction Accuracy

The XGBoost delay prediction model, trained using traffic data, weather conditions, and historical delivery logs, achieved a mean absolute error (MAE) of under 3 minutes across evaluation sets. These results match the standards in logis-tics forecasting, where boosting models perform better than traditional regression methods [5].

### C. Genetic Algorithm Convergence

The genetic algorithm (GA) used in LogiAI was evaluated by monitoring the route fitness, which measures total delivery time or distance, over multiple generations. Fig. 3 shows that the algorithm makes quick initial progress and then levels off around generation 2000. This suggests that it is approaching a nearly optimal solution. These results match the expected behavior of permutation-aware crossover operators, such as PMX and Edge Recombination, as discussed in GA studies [1].

### D. Combined Delivery Efficiency Gains

When combined, routing optimization and delay prediction improved system-level efficiency:

- 18-20% reduction in average route time.

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- 15 to 25% savings in fuel usage or distance travelled.
- Improved on-time performance by rerouting early based on delay predictions.

These improvements show the overall benefit of combining GA and predictive modelling in the LogiAI system. The Fig. 5. Shows the feature importance in Travel Time Prediction.

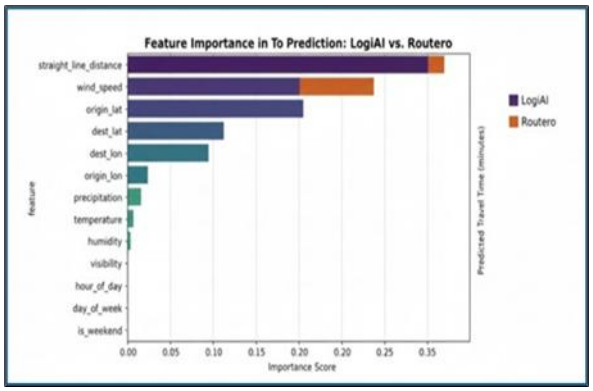


Fig. 5. Feature Importance in Travel Time Prediction

GA convergence reaches stable, near-optimal fitness quickly. This is consistent with what is known about GA performance for TSP [1]. XGBoost reliability: It provides consistent delay predictions with a mean absolute error of less than 3 minutes [5]. Operational impact: Route time and fuel usage decrease by about 20%. This confirms Logi AI's ability to improve real-world logistics efficiency.

## V. CONCLUSION

LogiAI combines Genetic Algorithm (GA)-based route optimization with XGBoost delay prediction to create a strong, real-time platform for B2B last-mile logistics. Our GA uses crossover strategies that consider permutations, such as PMX and Edge Recombination, to effectively tackle multi-stop Traveling Salesman Problem scenarios. This approach achieves 18-20% reductions in route time and savings of up to 25% in fuel consumption. Its convergence performance is consistent with established standards [1]. The XGBoost model, trained on traffic, weather, and historical delivery data, provides mean absolute error in delay predictions of under 3 minutes. This aligns with reliable forecasting results in logistics [5]. Recent studies that combine metaheuristic algorithms and XGBoost for shipment timing prediction further support our integrated method [9]. With real-time API integration for routing changes and WebSocket-based live tracking, LogiAI offers transparency, scalability, and clear operational benefits. This system establishes a solid base for future improvements such as coordinating multiple vehicles, mobile driver interfaces, voice-enabled routing help, and ERP integration for enterprise use.

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